Examiner only

4. Dilute hydrochloric acid reacts with magnesium forming magnesium chloride and hydrogen gas.

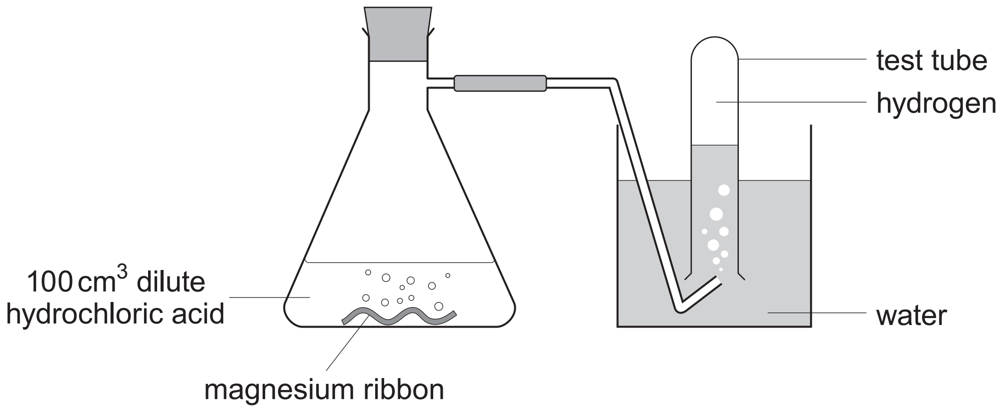
(a) Tick (✓) the box next to the correct equation for the reaction between magnesium and hydrochloric acid. [1]

$Mg + HCl \longrightarrow MgCl_2 + H$ ☐

$Mg + 2HCl \longrightarrow MgCl_2 + 2H$ ☐

$Mg + 2HCl \longrightarrow MgCl_2 + H_2$ ☐

(b) Osian wanted to find out how changing the concentration of the acid affects the rate of the reaction. He carried out five experiments at room temperature (20 °C). He added a 4 cm piece of magnesium ribbon to 100 cm^3 of hydrochloric acid of five different concentrations. He recorded the time it took to half-fill a test tube with gas.



His results are shown below.

Experiment	Acid concentration (%)	Time (s)
1	100	12
2	80	14
3	60	20
4	40	36
5	20	76

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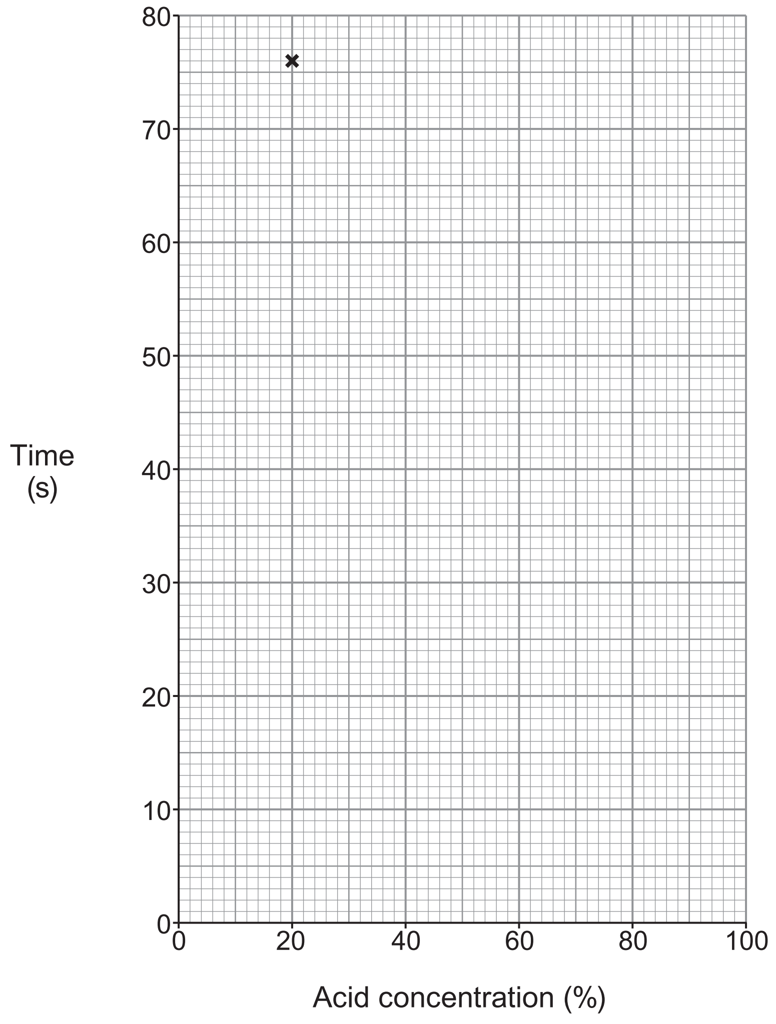
(i) Plot the acid concentration against time on the grid below and draw a suitable line.

One point has been plotted for you. [3]



(ii) Underline the correct word(s) in the brackets to complete the following sentences. [2]

As the acid concentration increases, the **time** to half-fill the test tube with gas **(increases / stays the same / decreases)**.

As the acid concentration increases, the **rate** of the reaction **(increases / stays the same / decreases)**.

Examiner only

(iii) Using your knowledge of particle theory, underline the correct words in the brackets to complete the following sentence. [2]

At a higher concentration, there are **(more / less / the same number of)** particles present so there will be **(an equal / a smaller / a greater)** chance of collision.

(iv) There are other ways the rate of the reaction can be changed.

Tick (✓) the **two** statements that correctly describe other ways the rate of reaction can be increased. [2]

Increasing the temperature of the acid	☐
Using a lump of magnesium	☐
Using a different apparatus	☐
Using magnesium powder	☐
Decreasing the temperature of the acid	☐

10